P5.JS + ML5.JS

Face tracking with faceMesh

How to find eyes, lips and the face outline — and draw with them

`face.leftEye` · `face.lips` · `face.faceOval`

STEP 1 • SETUP

Start faceMesh on the webcam

```
let faceMesh;
let video;
let faces = [];
function preload() {
  faceMesh = ml5.faceMesh({ maxFaces: 1 });
}
function setup() {
  createCanvas(640, 480);
  video = createCapture(VIDEO);
  video.size(640, 480);
  video.hide();
  faceMesh.detectStart(video, gotFaces);
}
```

```
function gotFaces(results) {
  faces = results;
}
function draw() {
  image(video, 0, 0);
  if (faces.length > 0) {
    let face = faces[0];
    // use face.leftEye etc.
  }
}
```

Add ml5 in **index.html**:
unpkg.com/ml5@1/dist/ml5.min.js

STEP 2 • WHAT YOU GET

The named parts of a face

Call	What it is
face.leftEye	Outline of the person's left eye
face.rightEye	Outline of the person's right eye
face.leftEyebrow / rightEyebrow	The two eyebrows
face.leftIris / rightIris	The irises (needs refineLandmarks: true)
face.lips	Outer and inner edge of the mouth
face.faceOval	Outline of the whole face, jaw to forehead
face.box	Rectangle round the face: xMin, yMin, width, height
face.keypoints	Every point in the mesh (468, more with irises)

STEP 3 • INSIDE A PART

Every part gives you the same values

Value	Meaning
.x .y	Top-left corner
.width	How wide it is
.height	How tall it is
.centerX	Middle, across
.centerY	Middle, down
.keypoints	Array of points, each with x, y, z

```
let eye = face.leftEye;
// box round the eye
noFill();
rect(eye.x, eye.y, eye.width, eye.height);
// dot in the middle
fill(255);
circle(eye.centerX, eye.centerY, 20);
// every point on the eye
for (let p of eye.keypoints) {
  circle(p.x, p.y, 4);
}
```

STEP 4 • TRY IT

Three quick recipes

Googly eyes

Circles sized to each eye

```
let L = face.leftEye, R = face.rightEye;
circle(L.centerX, L.centerY, L.width * 1.5);
circle(R.centerX, R.centerY, R.width * 1.5);
```

Is the mouth open?

Tall lips = open mouth

```
if (face.lips.height > 30) { // tune 30 for your camera
  text("Mouth open!", 20, 40);
}
```

Draw the mesh

A dot on every point

```
for (let p of face.keypoints) {
  point(p.x, p.y);
}
```

RECIPE • LIPS AS A SHAPE

Join the lip points with beginShape()

```
// point numbers that go round each lip outline, in order
let outer = [61, 185, 40, 39, 37, 0, 267, 269, 270, 409, 291, 375, 321, 405, 314, 17, 84, 181, 91, 146];
let inner = [78, 191, 80, 81, 82, 13, 312, 311, 310, 415, 308, 324, 318, 402, 317, 14, 87, 178, 88, 95];
```

```
function drawLoop(face, ids) {
  beginShape();
  for (let i of ids) {
    let p = face.keypoints[i];
    vertex(p.x, p.y);
  }
  endShape(CLOSE);
}
```

```
// inside draw(), after let face = faces[0]
noStroke();
fill(220, 30, 80);
drawLoop(face, outer); // lips
fill(40, 0, 15);
drawLoop(face, inner); // mouth
```

Complete sketch: drawing lips

```
let faceMesh, video, faces = [];
let outer = [61, 185, 40, 39, 37, 0, 267, 269, 270,
  409, 291, 375, 321, 405, 314, 17, 84, 181, 91, 146];
let inner = [78, 191, 80, 81, 82, 13, 312, 311, 310,
  415, 308, 324, 318, 402, 317, 14, 87, 178, 88, 95];

function preload() {
  faceMesh = ml5.faceMesh({ flipped: true });
}
function setup() {
  createCanvas(640, 480);
  video = createCapture(VIDEO, { flipped: true });
  video.size(640, 480);
  video.hide();
  faceMesh.detectStart(video, gotFaces);
}
function gotFaces(results) { faces = results; }
```

```
function draw() {
  image(video, 0, 0, width, height);
  if (faces.length > 0) {
    let face = faces[0];
    noStroke();
    fill(220, 30, 80);
    drawLoop(face, outer);
    fill(40, 0, 15);
    drawLoop(face, inner);
  }
}
function drawLoop(face, ids) {
  beginShape();
  for (let i of ids) {
    let p = face.keypoints[i];
    vertex(p.x, p.y);
  }
  endShape(CLOSE);
}
```

RECIPE • EYES, NOSE, FACE SHAPE

Same trick, different point lists

```
// face outline, forehead → right → chin → left
let oval = [10, 338, 297, 332, 284, 251, 389, 356, 454, 323, 361, 288, 397, 365, 379, 378, 400, 377,
  152, 148, 176, 149, 150, 136, 172, 58, 132, 93, 234, 127, 162, 21, 54, 103, 67, 109];
// eyes (the person's left and right)
let eyeL = [263, 466, 388, 387, 386, 385, 384, 398, 362, 382, 381, 380, 374, 373, 390, 249];
let eyeR = [33, 246, 161, 160, 159, 158, 157, 173, 133, 155, 154, 153, 145, 144, 163, 7];
// nose: top between eyes, one corner, bottom, other corner
let nose = [168, 98, 2, 327];
```

```
// in draw(), using drawLoop() from the lips slide
noFill(); stroke(255); strokeWeight(3);
drawLoop(face, oval);
fill(255); drawLoop(face, eyeL); drawLoop(face, eyeR);
fill(255, 180, 0); drawLoop(face, nose);
```

No nose outline in the mesh, so we join four nose landmarks into a kite.

Nose tip on its own: **face.keypoints[1]**

IF IT GOES WRONG

Common mistakes and fixes

Problem	Fix
Error: can’t read leftEye of undefined	No face yet — wrap your code in if (faces.length > 0)
leftEye shows up on the right	“Left” is the person’s left. Mirror with { flipped: true } in both ml5.faceMesh() and createCapture(VIDEO, …)
leftIris is missing	Turn it on: ml5.faceMesh({ refineLandmarks: true })
Only one face is found	Set maxFaces: 2 (or more) and loop over the faces array
Sketch is slow	Keep maxFaces low; call faceMesh.detectStop() when you no longer need tracking